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Name of Assignment: **Stage 3 - Prototyping**

Project Title: **Wireless blood pressure device for use in emergency care**

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Stage 3 Report - Wireless blood pressure device for use in emergency care

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Abstract

Emergency departments across Australia struggle with congestion resulting in poor patient outcomes such as extreme wait periods. Our solution was to create a device that monitors patients' blood pressure during the waiting period to shorten wait times and improve patient care. With our device patients will be more efficiently prioritised and triage nurses and doctors can review automated results for better insight towards care. Our product was built utilising relevant parts of the Withings Smart blood pressure monitor, which were used to create our prototype. The final prototype was was able to contract and release as appropriate, send and receive data wirelessly, and a backend solution created to establish a more complete product.

Keywords— IoT device, blood pressure, hospital, MQTT, vital sign, patient care, emergency room, triage system

I. INTRODUCTION

Patient volumes at emergency departments (EDs) are on the rise, which frequently leads to overcrowding, long wait times, and delayed patient care. This has led to an urgent demand for creative solutions that might improve the prioritisation of treatment for important cases and expedite patient management. The lengthy procedure of vital sign monitoring, which is a crucial factor in determining triage priority, is one of the main causes of ED delays [1]. Manually taking vital signs, such as blood pressure, using traditional methods is time-consuming and can cause delays, particularly in emergency departments with a high volume of patients or a shortage of staff. By enabling doctors to continuously follow changes in patient status, a dependable, real-time monitoring solution might greatly enhance patient outcomes by enabling quicker and more precise triage decisions.

Our team created a wireless blood pressure (BP) monitor especially for emergency care settings in order to overcome these difficulties. In order to prioritise essential patients according to the most recent data, this technology automates the collection of vital signs and changes triage scores in real-time [2]. Our device is intended to be worn constantly by patients, in contrast to traditional blood pressure cuffs, and provides frequent, precise readings without requiring manual intervention. Our technology performs well in comparison to other methods, including semi-automated monitoring stations or handheld blood pressure monitors, which still need occasional clinical assistance and don't always work well with current triage systems. Continuous monitoring systems have the potential to increase workflow efficiency and lower human error in emergency rooms, according to recent studies.

Even while our gadget appears promising, there are still a number of unknowns and hazards, especially with regard to data accuracy, device longevity in high-use situations, and possible interference with other wireless devices in emergency departments. We carried out a great deal of testing and prototyping to overcome these obstacles. Trials were carried out in simulated emergency departments to test real-time data transmission and integration with triage systems, and the accuracy, comfort, and battery life of the initial prototypes [3]. According to our first findings, this device can reliably automate the monitoring process; nevertheless, further testing is necessary to confirm its safety and dependability under a variety of settings and further reduce risks.

II. MATERIALS AND METHODS

A. Materials

The materials utilised in this study included Withings Smart blood pressure monitor, comprising an air compressor, pressure transducer, air valve, blood pressure cuff and an apple 30 pin cable. The following were also utilised; an Arduino board, ESP32, a MQTT broker, wires, LCD, breadboard, power supply, resistors, transistors, diodes and toolkit.

B. Methodology for Solution 1

The procedure began through inspiration from research of a past blood pressure project. The blood pressure monitor built through RishabhL provided instruction data extraction from an existing device [4]. The first step was to disassemble the Withings BP monitor and identify key areas of interest of the board, including air compressor, pressure sensor, and data transmission hardware. Due to Apple's proprietary connection, we sought a way to hijack data transmission from certain wires so that an Arduino may be able to read it. However, Apple security features and lack of documentation led to inability to extract and apply data from the working device.

C. Methodology for Solution 2

As it was not possible to follow through with our first solution, we followed our contingency plan to create a custom BP device using the parts salvaged from the Withings device. Research provided inspiration for the project regarding Lunes Cuatro instructions of blood pressure monitor [5]. Configuration of the cuff, pump and valve was provided. A small flat screwdriver among other tools supported deconstruction of the Withings device (appendix 1). The

blood pressure monitor core of Withings device was first to be deconstructed (appendix 3). Two screws wedged into the apple cable cover caused detachment. Protective covering was distorted to reveal the air compressor, air valve and the circuit board containing the pressure sensor (appendix 5). Desoldering techniques separated the pressure sensor from the old circuit board (appendix 9).

Deconstruction of pieces provided product codes; air valve 'KSV15C-3M DC 3V KOGE 110428C', air compressor 'BRA2430001MLP1 BPO2430004NA5F' and pressure sensor 'MPS-2107 12 K4402JE006' Codes supported research of relevant data sheets for project direction, expected performance of each component and appropriate current and voltage to prevent device damage. Post research, DC power lab supply was utilised at appropriate voltage to test functionality of air valve and compressor (appendix 12). Once functional, Arduino code was incorporated. An attempt was made to calibrate the sensor with Arduino use (appendix 18).

D. Methodology for Software

Initially to demonstrate triage solutions for emergency care nurses' computers the MQTT server was attempted. It was not possible to connect the ESP32 to this server, as Swinburne's Eduroam Wi-Fi would not establish a connection. In a second attempt, we used Bluetooth to pair the ESP32 to a phone to establish a proof of concept for wireless data transmission. The final product works through coding; '1' controls on/off functionality for the air compressor, '2' controls valve on/off and '3' starts the BP device as would be used in a hospital; the valve and the air compressor are turned on and the pressure inside the cuff is read until it reaches a predetermined PSI, then a final reading is taken the air compressor and valve are turned off and a timer started that continuously does this process every 30 seconds. This timer can be set as any value, for our demonstration we set it to 20 seconds however in a hospital setting this would be more appropriate at 5-minute intervals.

III. RESULTS

A. Device Prototype Results

Our blood pressure monitor (seen in Figure 1) reconstruction project has made substantial advancements, particularly with the effective integration of the pump and valve with an ESP32 microcontroller, facilitating dependable inflation and deflation cycles for precise results. Furthermore, we engineered and 3D-printed an extension for the ESP32, improving the device's structural integrity and overall functionality. Challenges emerged with the pressure sensor, which, although being functional, yields inconsistent readings, presumably owing to damage sustained during construction. Replacing this sensor is crucial for attaining the accuracy necessary for reliable blood pressure measurements.

A crucial insight from the prototype phase was the significance of effective power management for the ESP32. The ESP32, initially powered by a 9V battery, faced potential harm from high voltage, necessitating a transition to a more appropriate power supply. Upon transitioning to a power bank, we noted a marked enhancement in stability, with the ESP32 operating consistently. This modification has been demonstrated to be an optimal option, enhancing the device's capabilities with a safer and more efficient power supply. We have included a link to the demonstration of this device

working (https://drive.google.com/file/d/1l4OSgqnA-ZRg8TKmQoSgBvJlPm12vH_P/view?usp=sharing).

Figure 1. Device prototype

B. Software Prototype Results

This section will outline the results obtained from the software requirements stated earlier in the report. The prototype solution had software requirements of ease of setup, scalability of multiple devices, ensuring data integrity during transmission, ensuring data security, and ease of integration into the existing hospital network infrastructure. This led us to investigate avenues of using Bluetooth connectivity, database integration, and use of a Message Queuing Telemetry Transport (MQTT) server. It was decided to use a MQTT server, specifically the Mosquito broker, for our application for many reasons; low message overhead, low resource usage, low power consumption, and reliability of packet delivery [6]. The MQTT protocol is also used extensively using Internet of Things (IoT) devices so there are extensive guides for implementing a server and implementing communications between an ESP32 and a computer.

We initially planned on using a Mosquitto broker that was set up by Eric and hosted at Swinburne University via Eduroam. However, it was not possible to connect the ESP32 to this network without special permissions from the IT department, which were not received before the end of the project. However, we still decided to implement the MQTT server for the purpose of a prototype nurse station. The outcome of this implementation is seen in Figure 2 below. The end goal for this products software is it to be implemented into the already existing software that a triage nurse will use when checking in patients, so we focused more on the functionality rather than the design. As seen, this software currently has the capability to select between two BP devices, allocate patient information to the device, then initialise the device. We have also provided a link to a video showing the functionality of the software (https://drive.google.com/file/d/1Jztumge8hYaudhSU2b0eDcyA0M0p7BBr/view?usp=sharing).

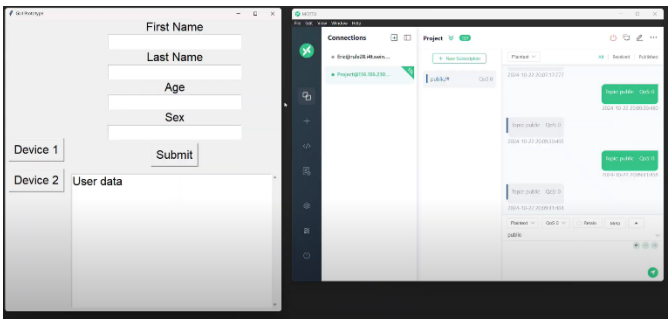

Figure 2. Software prototype

IV. Discussion

A. Device Prototype

Our blood pressure monitor re-creation prototype development showed promising functionality with effective integration of the pump, valve, and ESP32 microprocessor. While the pump and valve reacted precisely to orders, the pressure sensor posed a difficulty. Due to time constraints before the deadline, we could not repair the sensor, leading to inconsistent results that undermined measurement accuracy. In subsequent versions, this problem will be resolved by procuring a replacement sensor to guarantee accurate blood pressure measurements.

We intend to enhance the device's design to augment its functionality and aesthetics. A significant enhancement will be using a bespoke PCB to organize and optimize the electronics, minimizing external wiring and obtaining a more compact design. Should spatial limitations continue, we will re-engineer the 3D-printed extension to interlock firmly with the primary device, improving mobility and assembly efficiency. This strategy will enhance a more refined, market-ready aesthetic consistent with our streamlined, user-centric design objective.

We plan to enhance the power solution to improve the device's practicality for everyday usage. The existing power bank, albeit functional, introduces superfluous bulk. To resolve this, we want to incorporate a lithium battery, offering a more compact and convenient power source while ensuring reliable performance. This modification will render the gadget lighter and more portable, finalizing the necessary improvements to provide an optimum prototype for future uses.

B. Software Prototype

The software implementation for the BP device prototype was largely a success. A stated earlier, the main goals for this prototype were to have a system that can reliably and securely transmit data using a low power device and having patient data tied to any one device.

Various types IoT medical devices are already very well established, including those that collect health markers from patients such as heart rate, blood sugar, and BP [7,8]. The main concerns with these devices are how they will integrate into existing infrastructure, how to keep this information secure with so many different devices, and how this data can be useful for healthcare professionals and not contribute to information overload. Hao and Wang outline the need for dedicated infrastructure to support these new IoT medical devices, as the current infrastructure in many existing hospitals is unsuitable to support hundreds or thousands of these devices [9]. On top of this, it is critical to ensure these devices are properly secured, as the result of a unsecure device could be catastrophic not only to the patient, but could also provide a backdoor to the hospital network [10].

Our model does not aim to solve how integration into an existing infrastructure can occur, but it does provide security using TLS to ensure encryption of data between device and host. Whilst the prototype is operational, more research into network integration would be needed to ensure the success of the product.

C. Buisiness Strategy

The need for a solution to overcrowding in emergency departments, which leads to delays in care and poor patient outcomes. This issue is particularly critical for individuals showing early signs of clinical decline. Emergency departments, urgent care facilities, and hospitals, where healthcare systems are actively seeking tools that prioritize critically ill patients and expedite triage. The limitations in current solutions, such as the need for manual monitoring or reporting delays, which this device aims to overcome through continuous, real-time triage updates and monitoring.

The unique value of the device, featuring automated triage scoring, high accuracy, reliability, and seamless integration with hospital systems. Key differentiators include technological advantages like encrypted data transmission, automated notifications for out-of-range blood pressure readings, and continuous monitoring, ensuring rapid identification of critical patients.

The device will be marketed through direct hospital sales or a leasing option that minimizes upfront costs and includes regular maintenance. A subscription or service agreement is proposed to ensure continuous income through routine updates, calibrations, and technical support. Additionally, a data analytics service can provide hospitals with valuable insights into patient flow and triage efficiency, supporting improved care protocols.

Establishing key partnerships with medical organizations, medical device distributors, and potentially insurance providers to expedite product testing and regulatory approval processes. Regulatory alignment with standards such as ISO 13485 for medical devices enhances quality, safety, credibility, and market access [11]. Promotional efforts will use a multi-channel approach, highlighting the device's benefits through medical associations, conferences, and healthcare trade exhibitions.

Prioritize local markets initially, followed by international expansion upon regulatory approval and market validation. Future product line extensions could include versions for different healthcare settings, such as primary care offices or home monitoring for patients with chronic conditions. Technology upgrades will be informed by user feedback, potentially adding functionalities for additional vital signs or improvements in battery life and design.

Estimating production, marketing, and distribution costs to establish a competitive price that aligns with revenue goals and the device's value. Break-even and ROI projections consider the anticipated profitability timeline, including the expected ROI over the first few years, with specific milestones for reaching break-even based on partnerships, sales, or subscription uptake.

D. Future Work

For the second prototype for this BP device, we have identified three main areas of focus:

1. improved physical design,
2. accuracy of results, and
3. compatibility with infrastructure.

As we utilised two solutions for development, we first used an off the shelf device to try and complete our project. When this was no longer feasible, we continued to our second solution utilising the salvaged parts. Our second prototype would aim to use hand selected parts that can be then assembled in mass. As well as this, while the aluminium enclosure for the device suited the needs for the old product, this would not be feasible from a financial point of view but also a safety point of view having a large heavy object on multiple people. Our second prototype would utilise a custom design that would be more compact and made of more appropriate materials such as plastic or silicon.

As stated earlier, it was not possible to receive accurate results from the pressure sensor. In the future, ensuring that data is correctly read and displayed is vital to ensure the viability of this device. From our testing we see no possible issues that will arise from testing a sensor that is known to be working, and we expect it to show accurate results.

The infrastructure compatibility is the biggest roadblock for our device. It would be simple to alter the existing device to run over a different Mosquitto broker and establish communication between BP device and a computer. However, the issue lies with how we can test this device on an existing hospital network. As stated earlier, this is a well-documented issue that all IoT medical devices face when implementing into a hospital, and there is no one solution. Ongoing research in this field and close relations with hospitals is needed to ensure the success of all IoT medical devices.

V. Conclusion

In conclusion, overcrowding of hospitals in Australia is a critical concern. To address prolonged patient wait periods the designed blood pressure monitor device can be used by waiting individuals without direct supervision. This will reduce the waiting period and support triage nurses and doctors' insight into client health, as they are able to prioritise those in critical need. Ultimately, this device should eliminate extreme waiting during emergencies.

VI. References

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VII. Appendices

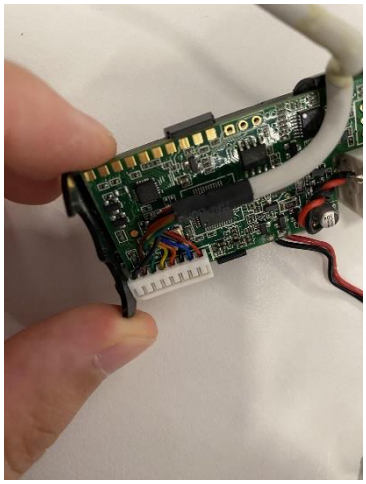
Appendix 1: The smart blood pressure monitor by Withings abstracted for device pieces: air pump, pressure transducer, air valve, blood pressure strap, transmission cord, tubes

Appendix 2: Withings brought blood pressure device unboxed and constructed

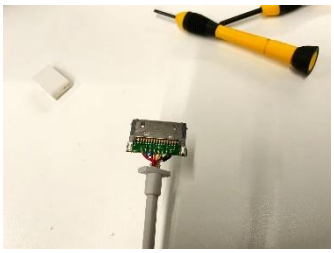
Appendix 3: BPM core before deconstruction

Appendix 4: BPM original placement

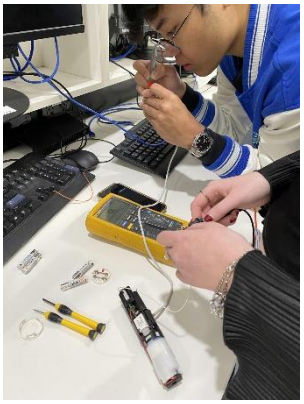
Appendix 5: BPM deconstructed for valuable parts



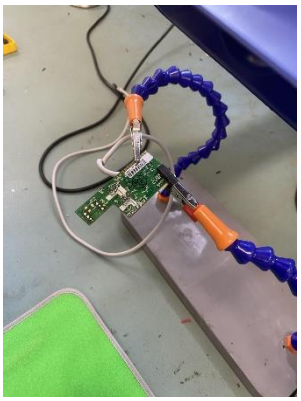
Appendix 6: PCB of original device



Appendix 7: Deconstructed iPhone/iPad cord obtained from Withings bp pack

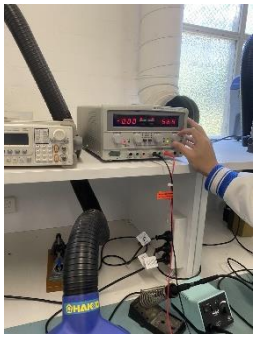


Appendix 8: Set up with use of Multimeter and lead probe use continuity in circuit

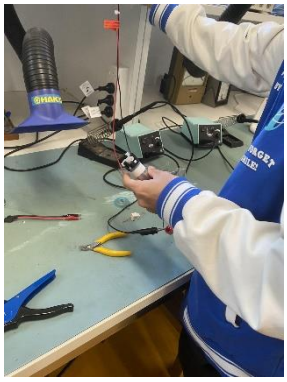


Appendix 9: process of desoldering pressure transducer from an old circuit

Appendix 10: Abstracted pressure transducer



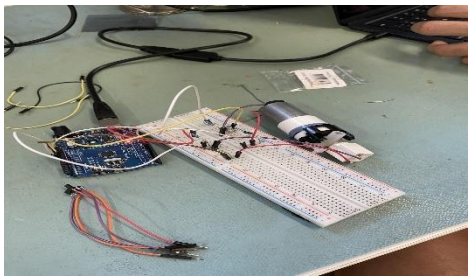
Appendix 11: DC power lab supply



Appendix 12: Set up with attempt to power air compressor

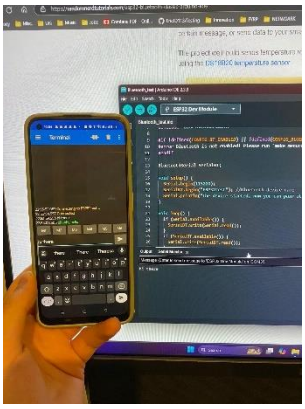
Appendix 13: Set up following Arduino coding

Appendix 14: Coding for testing pump functionality (on and off)

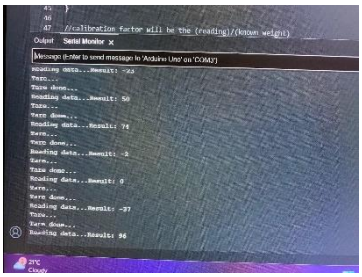


Appendix 15: Circuit used to turn pump on and off

Appendix 16: Setup for testing of pressure sensor

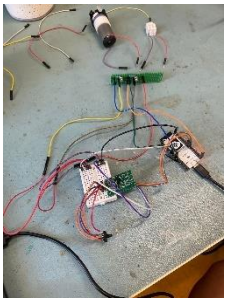


Appendix 17: Setup showing Bluetooth device working alongside code Arduino



Appendix 18: Calibration attempt

Appendix 19: Results of calibration



Appendix 20: Setup of integrated circuit

Appendix 21: Device encapsulated by 3D printed expansion model

Appendix 22: Final device setup